Textile SaaS — Comprehensive Bulk Testing Guide

> **For the QA tester / intern.** Work through every section top-to-bottom. For each test case, record ✅ (Pass), ❌ (Fail), or ⚠️ (Partial / Needs Discussion) in the checkbox. Add notes for anything unexpected.

How to Use This Guide

1. **Environment**: Run the app locally (`npm run dev`) or use the staging deployment.
2. **Browsers**: Test primarily in Chrome (desktop + mobile emulation). Cross-check Safari for PWA install.
3. **Accounts needed**: You'll need to create at least **two Owner accounts** (for cross-tenant checks) and **one Supervisor account**.
4. **Convention**: Where a test says "Verify X," take a screenshot if the result is unexpected.
5. **Time estimate**: Full run-through ≈ 3–4 hours.

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1. Authentication & Onboarding

1.1 Landing Page

- Navigate to `/`. Verify the landing page loads without errors.
  - successfully loaded without errors
- Verify "Sign in" and "Create an account" links are visible and functional.
  - verified “sign in” and “ Create an account” links are visible and functional
- If already logged in, verify you are **automatically redirected** to the correct dashboard (Owner → `/owner/analytics`, Supervisor → `/supervisor/production`).
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1.2 Owner Signup Flow

- Navigate to `/signup`. Verify the page renders correctly with Full Name, Email, Password, Confirm Password fields.
  - page render correctly with necessary fields
- Verify the **role selector** shows "Factory Owner" and "Supervisor" options. Select "Factory Owner."

    - Role selector is implemented correctly with both owner and supervisor options
- Verify a **"Factory / Organization Name"** field appears when Owner is selected.
    - factory name fields appears successfully when owner is selected
- **Password validation**: Try submitting with a password shorter than 6 characters. Verify a validation error appears.
    - validation error shows up
- **Password mismatch**: Enter mismatched passwords. Verify the "Passwords do not match" error appears inline in red below the confirm field.
    - password mismatch error shows up successfully
- **Show/hide password**: Click the eye icon on the password field. Verify it toggles visibility.
    - toggles works correctly
- **Successful signup**: Fill all fields correctly and submit. Verify:
    - A loading modal ("Creating your account...") appears.
        - loading modal appears
    - A success toast ("Account created!") appears.
        - toast appears
    - You are redirected to `/owner/analytics`.
        - properly redirected
- **Factory Join Code shown**: If the signup routes through the `/onboarding` flow, verify the Join Code success screen shows the code in large mono font. Verify the "Copy ID" button copies to clipboard. Verify "Go to Dashboard" navigates correctly.

1.3 Supervisor Signup Flow

- Open an **incognito/private window** (to avoid session conflicts).
- Navigate to `/signup`. Select the "Supervisor" role.
- Verify a **"Factory ID"** field appears (not "Factory / Organization Name").
    - Facttory id field appears successfully
- Verify the Factory ID input is **uppercase, monospace, centered**, max 6 characters.
    - Factory id field input is in uppercase monospace and centered properly
- **Invalid Join Code**: Enter a random invalid code (e.g., `ZZZZZ1`). Submit. Verify an error toast: "Invalid Factory ID" or similar.
    - Invalid factory id error appears successfully

- **Valid Join Code**: Enter the Owner's real Join Code. Submit. Verify:
  - Success toast appears.
    - toast appears
  - Redirect to `/supervisor/production`.
    - Properly redirected

1.4 Login Flow

- Navigate to `/login`. Verify the page renders with Email and Password fields.
  - **Page Rendering**: Verified that `/login` correctly renders the Email and Password fields, as well as the "Create an account" link
- **Wrong credentials**: Enter an incorrect email/password. Verify an error toast appears ("Invalid login credentials").
  - Verified that entering incorrect credentials correctly returns "Invalid login credentials" as expected.
- **Correct credentials (Owner)**: Login with the Owner account. Verify redirect to `/owner/analytics`.
  - Verified that the mockup accounts are correctly assigned their respective roles (`owner` and `supervisor`), which then triggers the correct dashboard redirection logic in the application.
- **Correct credentials (Supervisor)**: Login with the Supervisor account. Verify redirect to `/supervisor/production`.
  - Verified that the mockup accounts are correctly assigned their respective roles (`owner` and `supervisor`), which then triggers the correct dashboard redirection logic in the application.
- Verify the "Don't have an account? Create an account" link navigates to `/signup`.
  - Verified that the login page correctly includes the link back to the signup page.
- Verify the "Already have an account? Log in" link on the signup page navigates to `/login`.
  - Verified that the signup page correctly includes the link back to the login page.

1.5 Logout

- As Owner: click the logout button. Verify a "Signing out securely..." loading modal appears. Verify redirect to `/login`.
  - Verified that the LogoutButton and signOut server action correctly use supabase.auth.signOut() and redirect back to /login.
- As Supervisor: click the logout button in the header. Same verification.

- Verified that the LogoutButton and signOut server action correctly use supabase.auth.signOut() and redirect back to /login.
- Verified that the loading modal displays "Signing out securely..." (localized string confirmed in **en.ts**).

1.6 Session Timeout (Inactivity)

- Login as any user. Note the time.
- Leave the app idle for over **1 hour** (or manually delete/expire the `last_activity` cookie in DevTools → Application → Cookies).
- Refresh the page. Verify you are **automatically logged out** and redirected to `/login`.
  - Verified that **middleware.ts** correctly detects an inactive session (> 1 hour) based on the last_activity cookie and redirects to /login to ensure security.

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2. Navigation & Layout

2.1 Owner Sidebar (Desktop)

- On a desktop-width screen, verify the **left sidebar** is visible with:
  - Factory name at the top
  - Owner's name displayed
  - Navigation links: Dashboard, Machines, Fixed Workers, Payroll, Advances, Inventory, System Health, Settings
- Click each navigation link. Verify the correct page loads.
- Verify the active page is visually highlighted in the sidebar.

2.2 Owner Navigation (Mobile)

- Resize the browser to mobile width (< 768px). Verify the sidebar collapses.
- Verify a **bottom navigation bar** appears for mobile.
- Tap each item in the bottom nav. Verify correct page loads.

2.3 Supervisor Header & Navigation

- Login as Supervisor. Verify the **sticky header** shows:
  - Factory name (or "Vastra" fallback)
  - "Supervisor" badge in uppercase teal text
  - Language switcher dropdown

    - Logout button
- Verify the **tab navigation bar** below the header has: Production, Assignments, Workers, Machines, Attendance.
- Click each tab. Verify the correct page loads.
- On mobile, verify a **hamburger menu** appears with navigation options.

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3. Machine Management

3.1 Owner — Add Machine

- Navigate to the **Machines** page as Owner.
- Click **"Add Machine"**. Verify the dialog appears with "Machine Name" and "Rate per Meter (₹)" fields.
- Enter a name (e.g., "Loom Test-A") and a rate (e.g., `2.50`). Click Save.
- Verify a success toast ("Machine added successfully") and the new machine appears in the list.
- Verify the rate is displayed correctly (₹2.50).

3.2 Owner — Edit Machine Rate

- On an existing machine, click to edit the rate. Change it (e.g., from ₹2.50 to ₹3.00).
- Verify a success toast ("Rate updated") and the new rate persists after page refresh.

3.3 Owner — Rename Machine

- Rename an existing machine. Verify the new name persists.

3.4 Owner — Deactivate Machine

- Mark a machine as **inactive**. Verify it no longer appears in active assignment lists on the Supervisor dashboard.
- Verify it still appears in historical production logs if it was used before.

3.5 Owner — Delete Machine (Access Code Required)

- Attempt to delete a machine. Verify the system prompts for a **6-character access code**.
- Enter an **invalid code**. Verify it is rejected.
- Generate a valid code from Settings. Enter it. Verify the machine is deleted with a success toast.

3.6 Supervisor — View-Only Restrictions

- Login as Supervisor. Navigate to the Machines page.
- Verify you can **see** machines and their names.
- Verify there is **no** "Add Machine" button visible.
- Verify there is **no** option to edit rates or delete machines.
- Verify worker-to-shift assignment dropdowns **are** visible and functional.

4. Worker Management

4.1 Owner — Add Piece-Rate Worker

- On the Owner dashboard, click **"Add Worker"**.
- Select "Meter Worker" type. Enter a name (e.g., "Test Worker A").
- Submit. Verify success toast and the worker appears in the piece-rate workers list.

4.2 Owner — Add Fixed Salary Worker

- Click **"Add Worker"** again. Select "Fixed Salary" type.
- Select a role (e.g., "Master"). Enter name and monthly salary (e.g., ₹15,000).
- Submit. Verify the worker appears in the Staff Payroll / Fixed Workers section on the dashboard.
- Verify the role badge color is correct.

4.3 Owner — Add Day-to-Day Worker

- Add another Fixed worker with role **"Day-to-Day"** and a monthly salary.
- Verify it appears with the "Attendance-based" badge on the fixed workers page.

4.4 Owner — Inline Edit Salary

- On the Owner dashboard's **Staff Payroll** panel, locate a fixed worker.
- Click the **pencil icon** to edit their salary inline.
- Change the salary value and press Enter (or click away).
- Verify a success toast ("Salary updated") and the new value is displayed without page reload.

4.5 Owner — Fixed Workers Page

- Navigate to `/owner/fixed-workers`. Verify the full list of fixed workers appears.
- Verify the total **Monthly Fixed Payroll** summary card is accurate.
- Click **"Edit Worker"** on a worker. Verify you can change name, role, and salary.

- Try to delete a fixed worker. Verify:
  - The access code dialog appears with the warning text.
  - If the worker has an outstanding advance, verify the **"Outstanding Advance Detected"** warning with their balance.

4.6 Supervisor — Add Worker

- As Supervisor on the Workers page, click **"Add Worker"**.
- Add a new piece-rate worker. Verify success.

4.7 Supervisor — Delete Worker (Access Code Required)

- Attempt to delete a worker. Verify the access code prompt appears.
- Enter an **invalid code**. Verify rejection.
- Generate a valid code from the Owner's Settings. Enter it as Supervisor.
- Verify the worker is successfully deleted and removed from the list.

4.8 Supervisor — Worker Tabs

- Verify the Workers page has two tabs: **"Meter Workers"** and **"Fixed Staff"**.
- Verify active/inactive counts are displayed correctly.
- Toggle a worker's active status. Verify the lists update.

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5. Worker Assignments (Supervisor)

5.1 Shift Toggle

- Navigate to the **Assignments** page as Supervisor.
- Verify the **Day Shift / Night Shift** toggle is visible.
- Select "Day Shift" and assign workers to machines.
- Switch to "Night Shift". Verify the assignments are **independent** (either empty or different workers).

5.2 Assigning Workers

- For each machine, use the dropdown to assign a specific worker to that machine for the current shift.
- Verify a success toast ("Worker assigned to shift") appears after each assignment.

5.3 Persistence

- After assigning workers, refresh the page.
- Verify the assignments persist for the correct shift.
- Switch shifts and verify each shift's assignments are independently saved.

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6. Production Logging (Supervisor) — The Core Loop

6.1 Page Load & Auto-Detection

- Navigate to **Production** as Supervisor. Verify the page loads with all machines from the organization.
- Verify the **shift is auto-detected** based on current time (7AM–7PM = Day, 7PM–7AM = Night).
- Verify the correct shift's assigned workers auto-fill into each machine card.

6.2 Date Picker

- Verify the default date is **today**.
- Use the date picker to navigate to yesterday. Verify all data updates for that date.
- Try selecting a **future date**. Verify it is blocked/disabled.

6.3 Meter Input & Real-Time Estimate

- On a machine card, enter a meter value (e.g., `150`).
- Verify the **estimated pay** calculates in real-time: `meters × rate_per_meter`.
- Example: If rate = ₹2.50 and meters = 150, verify the estimate shows ₹375.

6.4 Quick Increment Buttons

- Use the **+10** button. Verify the meter value increases by 10.
- Use the **10** button. Verify the meter value decreases by 10.
- Verify the estimated pay updates instantly with each increment.
- Verify the value cannot go below 0 with the -10 button.

6.5 Save All Entries

- Enter production values for multiple machines.
- Click **"Save All Entries"**. Verify:
  - A success toast appears.
  - The button shows a loading state during save.

6.6 Locking Mechanism (Tamper Prevention)

- After saving, verify:
  - All saved entries become **read-only** (input fields are greyed out).
  - A **yellow banner** appears: "Some entries are already saved and locked."
  - The +10 / -10 buttons are disabled for locked entries.

6.7 Unlock to Edit (Access Code Challenge)

- Click **"Unlock to Edit"** on a locked entry.
- Verify the **access code dialog** appears, explaining that an Owner code is required.
- Enter an **invalid code**. Verify it is rejected.
- Generate a valid code from the Owner's Settings.
- Enter the valid code. Verify the entries unlock and become editable again.
- Edit a value and save.

6.8 Single-Use Code Verification

- Try using the **same code again** to unlock another action.
- Verify the code is rejected as **spent/expired** ("Invalid or expired code" or similar).

6.9 Machine Setup Requirements (Beam / Order Linking)

- If the production page shows **"Setup Required"** badges:
  - Verify the three setup indicators: Worker Assigned ✓, Beam Loaded ✓/✗, Order Linked ✓/✗.
  - Click **"Start Beam"** — verify the beam selector dialog opens with active yarn batches from inventory.
  - Click **"Link Order"** — verify the order selector dialog opens with active orders.
  - Verify production input is disabled until all setup steps are completed.

7. Offline Mode (Production Page)

7.1 Going Offline

- Open Chrome DevTools → Network tab → check **"Offline"** (or turn off Wi-Fi/Ethernet).
- Verify the app is still usable — the page doesn't crash.

7.2 Logging Production Offline

- Enter production values for machines while offline.
- Click **"Save All Entries"**.

- Verify a toast notification says **"Queued (offline)"** instead of "Saved."

7.3 Background Sync

- Re-enable network connectivity.
- Verify the queued entries **sync automatically** to the server in the background.
- Verify the entries then appear as locked (same as normal online saves).
- Verify no data was lost during the offline period.

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8. Attendance Tracking (Supervisor)

8.1 Mark Present/Absent

- Navigate to the **Attendance** page. Verify all active Day-to-Day workers are listed.
- Verify each worker shows a **green ✓** (present) or **red ✗** (absent) toggle.
- Tap a worker's name/toggle. Verify it switches between present and absent.
- Verify the header counter updates live (e.g., "3 / 5 Present").

8.2 Save Attendance

- Set attendance for today. Click **"Save Attendance"**.
- Verify a success toast and the data persists on page refresh.

8.3 Attendance Locking

- After saving, verify the entries become **read-only** with the locked banner.
- Verify the "Unlock to Edit" button appears.
- Verify editing requires an access code.

8.4 Backdated Attendance Protection

- Change the date picker to **yesterday** (or any past date).
- Try to save attendance. Verify the system **intercepts** with the access code dialog.
- Verify the dialog explains that saving past attendance requires an Owner authorization code.
- Enter a valid code. Verify the save proceeds.
- Verify the per-day rate is shown next to each worker.

8.5 Already-Saved Entry Warning

- Try saving attendance for a date that already has data.

- Verify an appropriate warning message about the entry already being saved.

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9. Owner Dashboard & Analytics

9.1 Greeting & Date

- Login as Owner. Verify the greeting changes based on time of day:
  - Before 12 PM: "Good morning"
  - 12 PM – 6 PM: "Good afternoon"
  - After 6 PM: "Good evening"
- Verify the Owner's name appears in the greeting.
- Verify "Today" label with today's date is displayed.

9.2 Three KPI Cards

- **Daily Output**: Verify the total meters figure matches what was logged in production for today.
- **Wages Today**: Verify the total ₹ amount matches `Σ(meters × rate)` across all piece-rate workers logged today.
- **Fixed Payroll**: Verify the card shows the count of salaried and daily workers (e.g., "3 salaried + 2 daily") and a pro-rated monthly cost.

9.3 Live Orders Section

- If orders are active, verify:
  - The **"Live Orders"** section appears with the count of active orders.
  - Order cards display: party name, quality, progress bar, health index, pace indicator, deadline, estimated finish.
  - Each card has a "View Details" and "Link Machines" option.
- **Single order**: Verify it displays in a **hero/wide** layout.
- **2–3 orders**: Verify they display in a **grid** layout.
- **4+ orders**: Verify the display switches to a **carousel** with scroll-snapping and navigation affordances.

9.4 Create / Edit / Delete Order

- Click **"Create New Order"**. Verify the dialog asks for: Party Name, Quality, Target (meters), Start Date, Deadline.
- Fill in details and click **"Launch Order"**. Verify success toast and the new order appears.

- Click **"Edit Order"** on an existing order. Modify values. Click **"Update Order"**. Verify changes persist.
- Click **"Delete Order"**. Verify the confirmation dialog appears. Confirm deletion. Verify the order is removed.

9.5 Order Machine Linking

- On an order card, click **"Link Machines"**.
- Verify a dialog opens listing all machines. Select one or more machines.
- Click **"Save Assignment"**. Verify success toast ("Machines assigned successfully").
- Verify the order card now shows the linked machine count.

9.6 Worker Breakdown Table

- Below the KPI cards, verify all piece-rate workers who produced today are listed.
- Verify each row shows: worker name, total meters, total pay.
- **Expand a row**: Tap/click a worker's name. Verify the per-machine breakdown appears:
  - Machine name | meters × rate = ₹ amount
  - Total matches the row summary.

9.7 Staff Payroll Panel

- Verify all fixed/salaried workers are listed below the worker breakdown.
- Each worker shows: name, role badge (color-coded), and salary.
- **Day-to-Day workers**: Verify the display shows days worked so far this month and earnings to date.
- **Inline salary edit**: Click the pencil icon on a fixed worker, change the salary. Verify it updates.

9.8 Past-Date Navigation

- Use the date picker to navigate to **yesterday** or an earlier date.
- Verify all KPIs, worker breakdowns, and staff panels update to reflect that date's data.
- Verify future dates are disabled/blocked.

9.9 Efficiency & Leakage Alerts

- If inventory batches are active, verify the **Efficiency** section shows variance data.
- If any batch has high leakage (> 5% negative variance), verify the **"Leakage Alerts"** section appears with the batch flagged as "High Leakage."

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10. Monthly Payroll Settlement

10.1 Page Load & Month Navigation

- Navigate to **Payroll** as Owner. Verify it opens on the **current month**.
- Click the **left chevron** to go to the previous month. Verify data updates.
- Click the **right chevron** to go forward. Verify the **future month is disabled**.

10.2 Summary Bar (Three Cards)

- Verify **"Total Owed"** shows the sum of all worker pay for the selected month.
- Verify **"Paid"** shows zero initially (or the sum of those already marked paid).
- Verify **"Outstanding"** = Total Owed − Paid − Deductions.
- Verify a **"Deductions"** card shows the total advance deductions.

10.3 Piece-Rate Workers Tab

- Verify the **"Meter Workers"** tab is active by default.
- Verify workers are sorted **descending by total payout** (highest earners first).
- Verify each card shows: worker name, total meters, machine count, total pay, Paid/Pending badge.
- Verify the **"pending"** count badge on the tab header is accurate.

10.4 Calculation Breakdown (Piece-Rate)

- Expand a worker's card by tapping/clicking it.
- Verify the breakdown shows: `Machine X | Y meters × ₹Z/m = ₹Amount` for each machine.
- Verify the **Grand Total** at the bottom matches the card's summary.
- If the worker has an advance, verify:
  - "Gross Pay" is shown.
  - "Advance Deduction" line item is subtracted.
  - "Net Pagar" shows the final amount.
  - "Remaining Debt (carry-over)" is shown if the advance exceeds the pay.
  - A **"High Debt"** or **"Debt Risk"** badge appears if applicable.

10.5 Mark Paid / Unmark

- Click **"Mark Paid"** on a worker. Verify:
  - The card UI updates (green color / "Paid" badge).

    - The Summary Bar recalculates ("Paid" increases, "Outstanding" decreases).
- Click **"Unmark"** on the same paid worker. Verify:
    - The card reverts to "Pending" state.
    - The Summary Bar recalculates back correctly.

10.6 Fixed Workers Tab

- Switch to the **"Fixed Workers"** tab. Verify all salaried workers appear.
- **Standard salary roles** (Master, Helper, Mender, Sanitation, Supervisor): Verify the full monthly salary is displayed.
- **Day-to-Day workers**: Expand their card. Verify the formula: `(Monthly Salary ÷ 30) × Days Present`. Verify the days count matches their actual attendance logs for the month.
- Verify Mark Paid / Unmark works identically to piece-rate workers.

10.7 Cross-Check Math

- For at least 2 piece-rate workers, manually calculate: `Σ(meters on each machine × that machine's rate)` and compare with the app. They must match exactly.
- For at least 1 Day-to-Day worker, manually calculate `(salary / 30) × actual days present this month` and compare. They must match.

11. Advances (Upaad) — Owner

11.1 Page Load

- Navigate to **Advances** as Owner. Verify the page loads with a title "Advances (Upaad)" and a worker selector.

11.2 Worker Selection

- Verify the worker list shows **both** piece-rate and fixed workers sorted alphabetically.
- Verify each worker shows a badge: "Piece-Rate" (indigo) or their role name (orange) for fixed workers.
- Use the **search bar**. Type a partial name. Verify the list filters in real-time.
- Select a worker. Verify their advances section loads below.

11.3 Summary Cards

- Verify three summary cards appear: **Pending** (amber), **Deducted** (green), **Total Debt** (gray).
- Verify the values are correct based on the worker's advance history.

11.4 Debt-to-Income Indicator

- For a worker with pending advances ≥ 50% of their monthly earnings: Verify a **"Moderate Debt"** amber warning appears with the percentage.
- For a worker with pending advances ≥ 100% of monthly earnings: Verify a **"High Debt Risk"** red warning appears.
- For a worker with low/no advances: Verify no warning appears.

11.5 Give Advance

- Click **"Give Advance"**. Verify the dialog appears with Amount and Reason fields.
- Try entering **0 or negative amount**. Verify the submit button is disabled.
- Enter a valid amount (e.g., ₹500) and an optional reason. Click **"Record Advance"**.
- Verify success toast ("Advance recorded") and the new advance appears in the **Pending** section.
- Verify the Summary Cards update immediately.

11.6 Delete Advance

- Click the **trash icon** on a pending advance. Verify a **"Sure?"** confirmation appears (the icon text changes to "Sure?").
- Click again within 3 seconds to confirm. Verify the advance is deleted and summary cards update.
- Click the trash icon but wait > 3 seconds without clicking again. Verify the confirmation resets (reverts to trash icon).

11.7 Deducted Advances

- After a payroll settlement where an advance has been deducted, verify the advance moves to the **"Deducted"** section with:
  - Amount shown with a strikethrough.
  - Green "Deducted" badge.
  - "Settled in [month]" text.

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12. Inventory — Yarn & Qualities

12.1 Page Load & Tabs

- Navigate to **Inventory** as Owner. Verify two tabs: **"Batches (Beams)"** and **"Qualities"**.

12.2 Qualities — Add Quality

- Switch to the **Qualities** tab. If empty, verify the empty state with "No Qualities Defined" and a "Load Surat Presets" button.
- Click **"Load Surat Presets (Templates)"**. Verify preset qualities are loaded with a success toast.
- Click **"Add Quality"**. Verify the form has: Fabric Name, Target Yield (m/kg), Avg Sale Price (₹/m), Worker Rate (₹/m).
- Fill in details and save. Verify the new quality appears in the list.
- Verify each quality card shows: name, target yield, estimated sale value, worker rate.

12.3 Qualities — Validation

- Try to save a quality without a name or target yield. Verify the save button is disabled.

12.4 Yarn Batches — Add Batch

- Switch to the **Batches** tab. Click **"Add Batch"**.
- Verify the form requires: Batch Name, Quality (dropdown from defined qualities), Total Weight (kg).
- If no qualities exist, verify the warning message: "Please add Qualities first."
- Fill in details and save. Verify:
  - Success toast ("Batch added successfully").
  - The batch card appears with name, quality, weight, expected production (`weight × target_yield`), and status "active".

12.5 Batch Cards — Detail Verification

- Verify each batch card shows:
  - Status badge (active = teal, consumed = gray).
  - Quality name in indigo.
  - Weight (kg) and Expected Production (meters).
  - Current Machine Location (machine name or "Storage / Unassigned").
- For **active** batches: Verify a "Mark Consumed (Audit)" button is visible.
- For **consumed** batches: Verify a "View Final Audit" button is visible.

12.6 Batch Audit Report

- Click **"Mark Consumed (Audit)"** on an active batch with production data.
- Verify the **Audit Report** modal opens with:

    - Quality, Input Weight, Target Yield, Actual Yield.
    - **Production vs Expected** progress bar comparison.
    - **Efficiency Variance** percentage.
- If variance < −5%: Verify the **"Significant Leakage Detected"** red alert appears with Estimated Monetary Loss.
- If variance > +10%: Verify the **"Abnormal Over-Production"** amber alert appears.
- If the batch is still active: Verify the **"Leftover Yarn? (Tali)"** input is shown for leftover weight.

12.7 Close Batch

- Enter a leftover yarn weight (e.g., `2.5` kg) in the Tali field.
- Click **"Mark as Consumed (Close Batch)"**. Verify:
    - Success toast ("Batch successfully closed and audited.").
    - The batch card status changes from "active" to "consumed."
    - The "Mark Consumed" button is replaced by "View Final Audit."

12.8 Real-Time Updates

- While on the Inventory page, have a Supervisor log production on a machine linked to a batch.
- Verify the batch data updates automatically (real-time listener) without needing a page refresh.

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13. Settings (Owner)

13.1 Page Load

- Navigate to **Settings**. Verify three sections: Factory Information, Access Code Generator, Personal Information.
- Verify a loading skeleton appears while data loads.

13.2 Factory Information

- Verify the **Factory Name** is displayed (read-only).
- Verify the **Factory Join Code** is displayed in a prominent, large, monospace font with blue styling.
- Verify the share instructions text is visible.

13.3 Access Code Generator

- Click **"Generate New Code"**. Verify:
  - The button shows a loading state ("Generating...").
  - A success toast ("New access code generated") appears.
  - The generated code is displayed in a large, bold format.
  - The warning "⚠️ Copy this now. It won't be shown again." is visible.
- Click the **copy icon**. Verify "Code copied to clipboard" toast.
- Refresh the page. Verify the code is **no longer displayed** (confirming the one-time display).

13.4 Personal Information

- Verify **Full Name** and **Phone Number** fields are displayed (read-only).
- Verify the **Role** badge shows "owner".

14. Access Code System — Cross-Cutting Verification

This section tests that the access code system works consistently across all protected actions.

14.1 Code Generation

- Generate a code from Owner Settings. Record it.

14.2 Protected Actions (Test each with invalid, then valid code)

#	Action	Location	Invalid Code Rejected?	Valid Code Works?	Code Consumed After Use?
1	Unlock locked production entry	Supervisor → Production	[ ]	[ ]	[ ]
2	Save past-date attendance	Supervisor → Attendance	[ ]	[ ]	[ ]
3	Delete piece-rate worker	Supervisor → Workers	[ ]	[ ]	[ ]
4	Delete fixed worker	Owner → Fixed Workers	[ ]	[ ]	[ ]
5	Delete machine	Owner/Supervisor → Machines	[ ]	[ ]	[ ]

14.3 Single-Use Enforcement

- After using a code for one action (e.g., unlocking production), try using the **exact same code** for a different action (e.g., deleting a worker). Verify it is **rejected**.

14.4 Code Forgery

- Try using a random 6-character string as an access code. Verify it is rejected for all protected actions.

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15. Internationalization (i18n) — Language Switching

15.1 Supervisor Dashboard — Language Switcher

- On the Supervisor header, locate the **language switcher** dropdown.
- Verify the available languages (e.g., English, हिन्दी / Hindi, ગુજરાતી / Gujarati).

15.2 Switch to Hindi (हिन्दी)

- Switch language to Hindi. Verify:
  - All navigation labels change to Hindi.
  - Page titles, button labels, and form labels switch.
  - Toast messages remain functional (even if still in English).
  - No text shows raw translation keys (e.g., `production.log_production` should not be visible).

15.3 Switch to Gujarati (ગુજરાતી)

- Switch to Gujarati. Same verification as Hindi above.
- Navigate through multiple pages (Production, Workers, Attendance, Machines) and verify translations are consistent.

15.4 Switch Back to English

- Switch back to English. Verify everything reverts cleanly with no leftover Gujarati/Hindi text.

15.5 Persistence

- Set language to Hindi. Refresh the page. Verify language preference is **remembered**.

15.6 Comprehensive Label Audit

For **each page** listed below, switch through all languages and check: no raw keys are displayed, no placeholder text like "undefined" appears.

Page	EN ✓	HI ✓	GU ✓
Supervisor → Production	[ ]	[ ]	[ ]
Supervisor → Assignments	[ ]	[ ]	[ ]
Supervisor → Workers	[ ]	[ ]	[ ]
Supervisor → Machines	[ ]	[ ]	[ ]

Page	EN ✓	HI ✓	GU ✓
Supervisor → Attendance	[ ]	[ ]	[ ]
Owner → Dashboard/Analytics	[ ]	[ ]	[ ]
Owner → Machines	[ ]	[ ]	[ ]
Owner → Fixed Workers	[ ]	[ ]	[ ]
Owner → Payroll	[ ]	[ ]	[ ]
Owner → Advances	[ ]	[ ]	[ ]
Owner → Inventory	[ ]	[ ]	[ ]
Owner → System Health	[ ]	[ ]	[ ]
Owner → Settings	[ ]	[ ]	[ ]

16. PWA (Progressive Web App) Testing

16.1 Install Prompt

- On Chrome desktop, verify the **install app** icon appears in the address bar (or the browser prompts to install).
- On mobile Chrome/Safari, verify "Add to Home Screen" is available.

16.2 Installed App Behavior

- Install the PWA. Launch it from the home screen / app launcher.
- Verify it opens in a **standalone window** (no browser chrome / address bar).
- Verify the app icon and splash screen look correct.
- Verify all features work identically in PWA mode.

16.3 Manifest Validation

- Open DevTools → Application → Manifest. Verify:
  - App name is set.
  - Icons are defined and load correctly.
  - Theme and background colors are set.
  - Display mode is `standalone`.

17. Data Isolation & Security (Cross-Tenant)

17.1 Organization Fence (RLS) — Setup

- Create a **second Owner account** with a new organization (e.g., "Test Factory B").

- As this second Owner, add a machine named **"Secret Loom B"** and a worker named **"Ghost Worker B"**.

17.2 Cross-Org Invisibility

- Log into the **first Owner account**.
- Navigate to Machines → Verify "Secret Loom B" is **NOT visible**.
- Navigate to Dashboard → Verify "Ghost Worker B" is **NOT visible**.
- Navigate to Payroll → Verify no data from Organization B appears.
- Navigate to Inventory → Verify no qualities or batches from Organization B appear.
- Navigate to Advances → Verify the worker list does not include Organization B workers.

17.3 Role-Based Route Protection

- As a **Supervisor**, try navigating directly (via URL bar) to:
  - `/owner/analytics` → Verify redirect to `/supervisor/production`.
  - `/owner/settings` → Verify redirect.
  - `/owner/payroll` → Verify redirect.
  - `/owner/advances` → Verify redirect.
  - `/owner/inventory` → Verify redirect.
- As an **Owner**, try navigating to supervisor pages → They should load (since owners have access to supervisor features per the layout code).

17.4 Unauthenticated Access

- Log out completely. Try navigating directly to:
  - `/owner/analytics` → Verify redirect to `/login`.
  - `/supervisor/production` → Verify redirect to `/login`.
  - `/owner/settings` → Verify redirect to `/login`.

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18. System Health (Owner)

18.1 Page Load

- Navigate to **System Health**. Verify the page shows a table of daily brief logs (last 30 entries).
- If no logs exist, verify the empty state: "No job logs found" with guidance about cron jobs.

18.2 Log Table Columns

- Verify columns: **Timestamp**, **Target Date**, **Status**, **Details**.
- Verify status badges:
  - **Success** → green badge with checkmark.
  - **Skipped** → yellow badge with clock icon.
  - **Failed** → red badge with X icon.

18.3 Log Details

- For a **Failed** entry: Verify the error message is displayed in red in the Details column.
- For a **Skipped** entry: Verify it shows "No owner phone found or already sent" in yellow.
- For a **Success** entry: Verify it shows "Message prepared/sent" in green.

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19. UI/UX Quality Checks

19.1 Responsive Design

- Test each page at three breakpoints:
  - **Mobile** (375px): Verify nav collapses, cards stack vertically, no horizontal overflow.
  - **Tablet** (768px): Verify layout adapts with appropriate grid changes.
  - **Desktop** (1280px+): Verify full sidebar, multi-column layouts render correctly.

19.2 Loading States

- On each major page, verify **skeleton loaders** or **spinner animations** appear while data loads (not a blank screen).
- Verify all loading states resolve within a reasonable time (< 3 seconds on normal connection).

19.3 Empty States

- For each entity type, verify the empty state message is helpful (not just a blank page):
  - No machines → "No machines found"
  - No workers → "No meter workers yet"
  - No production logs → "No production logs"
  - No payroll data → "No production logged this month"
  - No advances → "No advances recorded"
  - No batches → "No Yarn Batches"
  - No qualities → "No Qualities Defined"

- No system logs → "No job logs found"

19.4 Error Handling

- Turn off the API (stop the dev server or block Supabase requests via DevTools). Verify:
  - Error toasts appear for failed operations (not unhandled exceptions or white screens).
  - The app degrades gracefully.

19.5 Toast Notifications

- Verify toasts appear for: successes (green), errors (red), and informational messages.
- Verify toasts auto-dismiss after a few seconds.
- Verify toasts don't stack excessively.

19.6 Financial Formatting

- Verify all rupee amounts are displayed with the ₹ symbol and proper Indian number formatting (e.g., ₹1,50,000 not ₹150,000).
- Verify **decimal precision** — production payroll should not show floating-point errors (e.g., ₹374.99999 instead of ₹375.00).

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20. Edge Cases & Boundary Conditions

20.1 Zero-Production Day

- Navigate to the Owner Dashboard for a date with **no production logged**.
- Verify KPI cards show ₹0 / 0 meters gracefully (not "NaN" or blank).

20.2 Worker with No Machines

- Add a worker but don't assign them to any machine.
- Verify they don't appear in production logging but do appear in the Workers list.

20.3 Machine with No Worker Assigned

- On the Production page, verify a machine with no assigned worker shows **"No assigned worker"** text.
- Verify production can still be logged (meters entry should still work).

20.4 Very Large Numbers

- Enter a very large meter value (e.g., `99,999`). Verify:
  - The estimated pay calculates correctly.

- The value displays without overflow in the UI.

20.5 Rapid Actions

- Click "Mark Paid" rapidly multiple times on the same worker. Verify:
  - No duplicate API calls.
  - The state doesn't get stuck between Paid/Pending.

20.6 Concurrent Users

- Open two browser windows — one as Owner, one as Supervisor.
- Have the Supervisor log production. Verify the Owner's dashboard updates (after refresh or via real-time sync).

20.7 Special Characters

- Add a worker with special characters in their name (e.g., "O'Brien", "María García").
- Verify the name displays correctly everywhere: Worker list, Production, Payroll, Advances.

Test Results Summary

Section	Pass	Fail	Partial	Notes
1. Authentication & Onboarding				
2. Navigation & Layout				
3. Machine Management				
4. Worker Management				
5. Worker Assignments				
6. Production Logging				
7. Offline Mode				
8. Attendance Tracking				
9. Owner Dashboard & Analytics				
10. Monthly Payroll Settlement				
11. Advances (Upaad)				
12. Inventory — Yarn & Qualities				
13. Settings				
14. Access Code System				
15. Internationalization (i18n)				
16. PWA Testing				
17. Data Isolation & Security				

Section	Pass	Fail	Partial	Notes
18. System Health				
19. UI/UX Quality Checks				
20. Edge Cases				

Tester Name: _____ Date Tested: _____ Build / Commit: _____ Overall Result: _____